Master Data Scientist Interview Question Bank

1. Python Fundamentals

Core Python

1. What are Python's main features?
2. Difference between list, tuple, set, and dictionary?
3. Mutable vs immutable objects?
4. What happens internally when you create a Python variable?
5. is vs ==?
6. Shallow copy vs deep copy?
7. What is Python's garbage collection?
8. What is reference counting?
9. What are namespaces?
10. Local vs global vs nonlocal variables?
11. What are *args and **kwargs?
12. What are lambda functions?
13. List comprehension vs generator expression?
14. What is an iterator?
15. What is a generator?
16. yield vs return?
17. What are decorators?
18. What are closures?
19. What is a context manager?
20. Explain with open(...).
21. What are dunder methods?
22. __str__ vs __repr__?
23. What is duck typing?
24. What is monkey patching?
25. What is method resolution order (MRO)?

Python OOP

26. Class vs object?
27. Instance variable vs class variable?
28. Instance method vs class method vs static method?
29. Explain encapsulation.
30. Explain abstraction.
31. Explain inheritance.
32. Single vs multiple vs multilevel inheritance?
33. What does super() do?
34. What is polymorphism?
35. Abstract class vs interface-like design in Python?
36. How do you implement private attributes?
37. What is property/decorator?
38. Composition vs inheritance?
39. When would you use composition?
40. Explain MRO with multiple inheritance.

---

2. Python Coding for Data Science

41. Reverse a string without using slicing.
42. Find duplicate elements in a list.
43. Find the first non-repeating character.
44. Count frequency of elements.
45. Find the second-largest number.
46. Find missing numbers from a sequence.
47. Remove duplicates while preserving order.
48. Flatten a nested list.
49. Merge two dictionaries.
50. Sort a dictionary by values.

51. Find common elements between two lists.
52. Implement your own map().
53. Implement your own filter().
54. Implement your own reduce().
55. Write Fibonacci using recursion.
56. Write Fibonacci iteratively.
57. Check whether a number is prime.
58. Find factorial.
59. Find palindrome.
60. Find anagrams.
61. Find top-K frequent elements.
62. Implement binary search.
63. Implement stack using Python.
64. Implement queue using Python.
65. Find time complexity of common Python operations.
66. Optimize nested loops.
67. How would you process a 20 GB CSV?
68. How would you reduce Python memory usage?
69. Why can vectorized NumPy code be faster than Python loops?
70. Multiprocessing vs multithreading?

---

3. NumPy

71. What is NumPy?
72. NumPy array vs Python list?
73. What is vectorization?
74. What is broadcasting?
75. Explain broadcasting with an example.
76. reshape() vs resize()?

77. flatten() vs ravel()?

78. view() vs copy()?

79. What is axis 0?

80. What is axis 1?

81. Explain matrix multiplication.

82. * vs @?

83. How does NumPy handle missing values?

84. What is NumPy memory layout?

85. Why are NumPy operations faster?

86. How would you calculate covariance using NumPy?

87. How would you normalize an array?

88. How would you calculate pairwise distances?

89. How would you detect NaN values?

90. How would you optimize a large NumPy computation?

---

4. Pandas

91. Series vs DataFrame?

92. loc vs iloc?

93. at vs iat?

94. merge() vs join() vs concat()?

95. Inner vs left vs right vs outer join?

96. groupby() internally does what?

97. agg() vs apply() vs transform()?

98. When would you use transform()?

99. How do you handle missing values?

100. dropna() vs fillna()?

101. Mean vs median imputation?

102. How do you detect duplicates?

103. How do you rename columns?
104. How do you change data types?
105. How do you handle inconsistent categorical values?
106. How do you detect outliers using Pandas?
107. How do you handle dates?
108. How do you extract year/month/day?
109. How do you calculate rolling averages?
110. How do you calculate cumulative values?
111. How do you find top-N values per group?
112. How do you identify duplicate records?
113. How do you handle a 10 GB dataset?
114. How do you optimize DataFrame memory?
115. Why is apply() sometimes slow?
116. How would you replace apply() with vectorization?
117. How do you perform feature engineering using Pandas?
118. How do you avoid chained assignment?
119. What is SettingWithCopyWarning?
120. How do you process data in chunks?

---

5. SQL

SQL remains a major part of Data Scientist interviews, particularly joins, aggregations, window functions, and practical data manipulation.

SQL Fundamentals

121. WHERE vs HAVING?
122. GROUP BY vs ORDER BY?
123. INNER JOIN vs LEFT JOIN?
124. UNION vs UNION ALL?
125. DELETE vs DROP vs TRUNCATE?